Self-Scaled Quasi-Newton Methods for Physics-Informed Neural Networks

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Thesis. Self-scaling stabilizes curvature information in stiff fourth-order PINN losses, enabling lower solution error than standard BFGS / L-BFGS-B—but only when paired with Adam warmup, residual resampling, and physics-based validation.

The Cahn–Hilliard Equation

$$\frac{\partial U}{\partial t} = D\,\nabla^2\!\left(\nabla^2 U + a_2 U + a_4 U^3\right)$$

Fourth-order phase-separation PDE with canonical $D{=}a_2{=}a_4{=}1$ on $[0,128]^2\times[0,20]$, periodic BCs, and smoothed random ICs $U_0\sim\mathcal{U}(-1,1)$ with $\sigma{=}2$. **Conservation & gradient-flow:** $\frac{d}{dt}\int_\Omega U\,d\mathbf{x}=0,\ \partial_t U = D\nabla^2\mu,\ \mu=\delta F/\delta U.$

Why Hard?

- Fourth-order derivatives $\nabla^4 U$ ($|\mathbf{k}|^4$).
- Cubic coupling $\nabla^2(U^3){=}6U|\nabla U|^2{+}3U^2\nabla^2 U$.
- Loss ill-conditioned: $\kappa(\nabla^2\mathcal{L})\gg 1$.

PINN Residual & Loss

$$f_\theta(t,\mathbf{x}) = \partial_t U_\theta - D\,\nabla^2\!\left(\nabla^2 U_\theta + a_2 U_\theta + a_4 U_\theta^3\right)$$

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N_f}\sum_i |f_\theta(\mathbf{z}_i)|^2}_{\mathcal{L}_{\mathrm{PDE}}} + 5\,\mathcal{L}_{\mathrm{IC}} + 5\,\mathcal{L}_{\mathrm{BC}}$$

Network U_θ: $[3,20{,}20{,}20{,}20,1]$ MLP, $\tanh$, $|\theta|{=}1{,}361$.

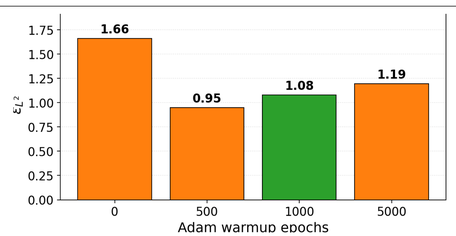
Self-Scaled Quasi-Newton

Build inverse-Hessian H_k from gradient differences; rescale by local curvature.

$$\tau_k \sim \frac{s_k^\top s_k}{y_k^\top s_k},\quad p_k{=}{-}H_k g_k,\quad H_{k+1}{=}\text{scaled-Broyden}(H_k, s_k, y_k; \tau_k, \varphi_k)$$

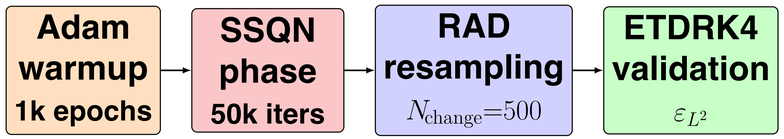
where $s_k{=}\theta_{k+1}{-}\theta_k,\ y_k{=}g_{k+1}{-}g_k$. **Variants:** BFGS, L-BFGS-B, SSBFGS-OL/AB, SSBroyden1, **SSBroyden2**.

Warmup Ablation



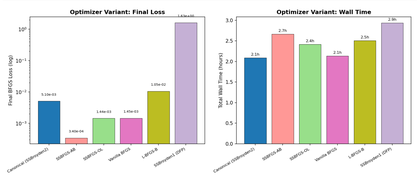
500–1000 Adam epochs stabilize secant pairs; zero warmup diverges.

Training Pipeline

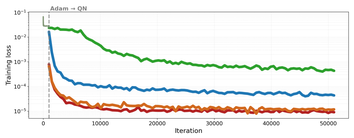


Experimental sweep: 6 optimizer families · 3 seeds/variant · 50k QN iters with ETDRK4 reference on $128{\times}128$; the report adds a 64^2 campaign where canonical SSBroyden2 reaches mean $\varepsilon_{L^2}=0.317$ and best seed 0.258.

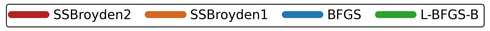
128^2 Optimizer Evidence



final ε_{L^2} ranking across variants



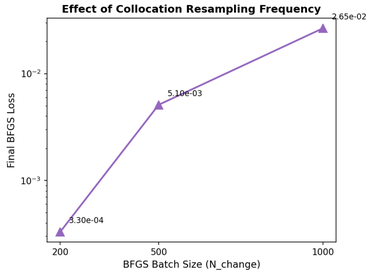
thicker BFGS-phase convergence traces



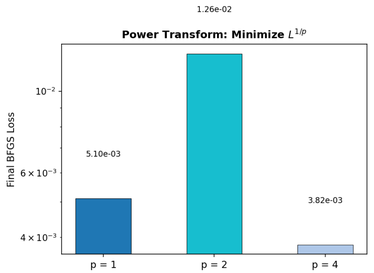
detached variant legend for poster readability

On $128{\times}128$, optimizer choice and resampling interact strongly: the best single-seed run is SSBroyden2 with $N_{\text{change}}=200$, achieving $\varepsilon_{L^2}=0.854$.

128^2 Ablations



frequent resampling helps

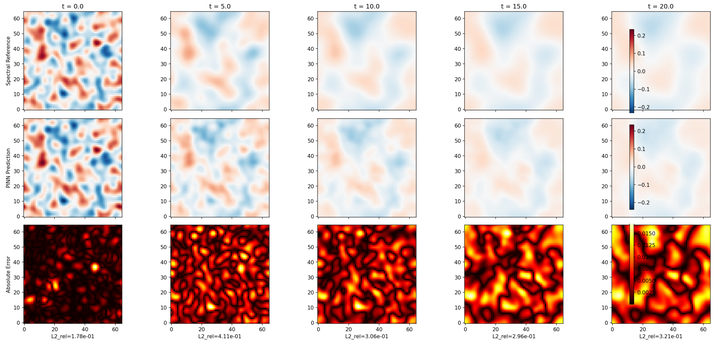


$p=4$ improves the loss geometry

The report finds that aggressive resampling and power transforms help on 128^2, but these same corrections are non-transferable on 64^2.

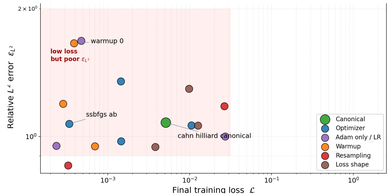
Best 64^2 Prediction vs. ETDRK4

Report result: canonical SSBroyden2 reaches mean $\varepsilon_{L^2}=0.317$ and best seed 0.258.

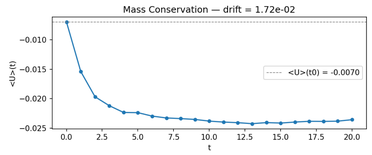


Rows show reference, PINN prediction, and absolute error for $t\in\{0,5,10,15,20\}$.

Validation Beyond Training Loss



low residual $\not\Rightarrow$ low ε_{L^2}



direct diagnostic from 'predictions/'

Validation must be physics-aware: compare against ETDRK4 and monitor conserved mass, not just the collocation residual.

Report Takeaways

1. **SSBroyden2** is strongest on both domains, but the gain is much larger on 64^2.
2. On 128^2, the best single-seed run is batch-200 SSBroyden2 with $\varepsilon_{L^2}=0.854$.
3. On 64^2, the simple canonical setup is already near-optimal; extra tweaks mostly hurt.
4. Power, RAD, and resampling corrections are **non-additive**: combined modifications interfere.